

A collection of historical artifacts is arranged on a light-colored surface. In the top left, a portion of a wooden chessboard with a checkered pattern and several chess pieces is visible. Below the chessboard, there are two medals: one with a red ribbon and a white star, and another with a blue ribbon and a white star. A pair of round-rimmed glasses with thin metal frames lies diagonally across the center. In the bottom left corner, a small, round, silver-colored compass is partially visible.

Distributed Rational Decision Making -- Tuomas Sandholm

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Evaluation Criteria

- ◆ Social Welfare (maximum)
- ◆ Pareto efficiency
- ◆ Individual Rationality
- ◆ Stability
- ◆ Computational efficiency
- ◆ Communication efficiency



Social Welfare

- ◆ Sum of all agent's payoffs or utilities



Pareto efficiency

- ◆ A solution x is Pareto efficient, if there is no other solution x' such that at least one agent is better off in x' than in x and no agent is worse off in x' than in x .
- ◆ Social welfare maximizing solutions are a subset of Pareto efficient ones.



Individual Rationality

- ◆ Agents are assumed to be self-interested
- ◆ Payoff in the solution in negotiation must not be less than the payoff the agent could get without participating in the negotiation.



Stability

- ◆ Dominant strategy: the strategy that an agent is better off no matter what strategies other agent use
- ◆ Nash equilibrium: no agent will leave the equilibrium alone
 - unique, multiple, no Nash equilibrium

- 
- ◆ Subgame perfect Nash equilibrium:
 - A Nash equilibrium that remains a Nash equilibrium in every subgame
 - also suffer existence and unique problem
 - ◆ Strong Nash equilibrium: No sub-group of agents will leave the equilibrium together (no collusive deviation)



Stability

- ◆ Prisoner's dilemma: A Nash solution that is not the Pareto optimal one.



Computational efficiency

- ◆ Prefer lowest computation overhead it find the solution



Communication efficiency

- ◆ Minimize the amount of communication that is required to converge on a desirable global solution.



Voting (Social Choice)

- ◆ An agent $i \in A$ has a asymmetric and transitive strict preference relation $>_i$ over a set of possible outcomes O .
- ◆ A social choice rule takes as input the agent's preference relations $(>_1, >_2, \dots, >_{|A|})$ and produces as output the social preferences denoted by a relation $>^*$.



Properties of a social choice rule

- ◆ A social preference ordering $>^*$ should exist for all possible inputs (individual preferences) [completeness]
- ◆ $>^*$ should be defined for every pair $o, o' \in O$
- ◆ $>^*$ should be asymmetric and transitive over O
- ◆ The outcome should be Pareto efficient: if $\forall i \in A, o >_i o'$ then $o >^* o'$ [Paretian]
- ◆ The scheme should be independent of irrelevant alternatives.
- ◆ No agent should be dictator in the sense that $o >_i o'$ implies $o >^* o'$ for all preferences of the other agents



Arrow's impossibility theorem

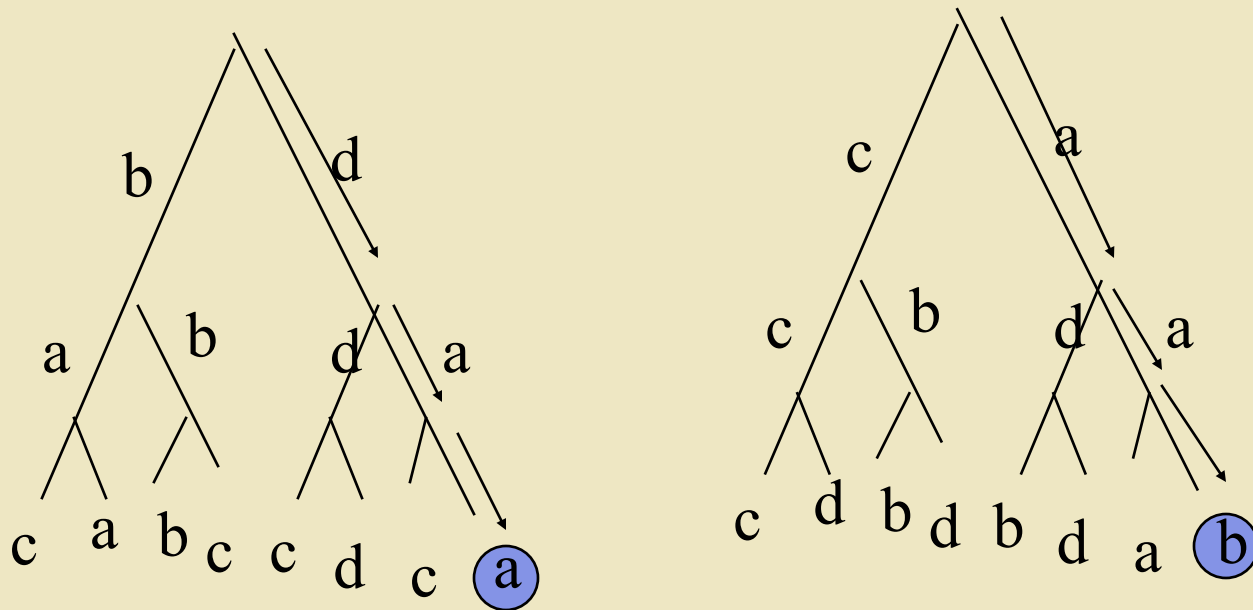
- ◆ No social choice rule satisfies all of these six conditions



Relaxation of the criteria

- ◆ Plurality protocol, Majority count: relax the third (symmetry)
- ◆ Binary protocol: irrelevant alternatives might alter the social choice, the agenda is important.

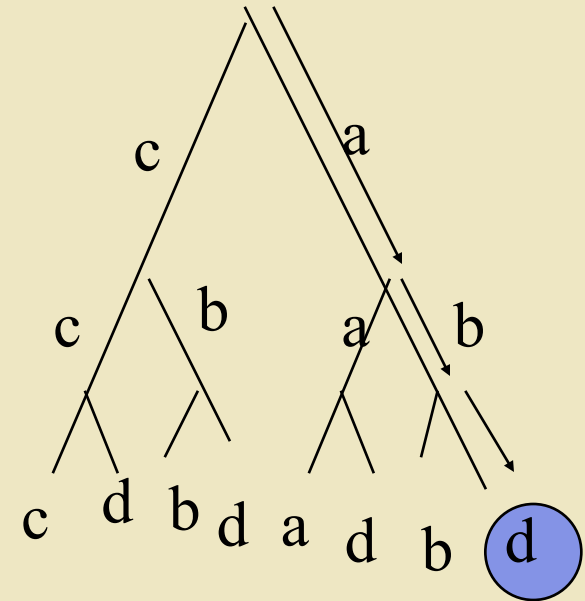
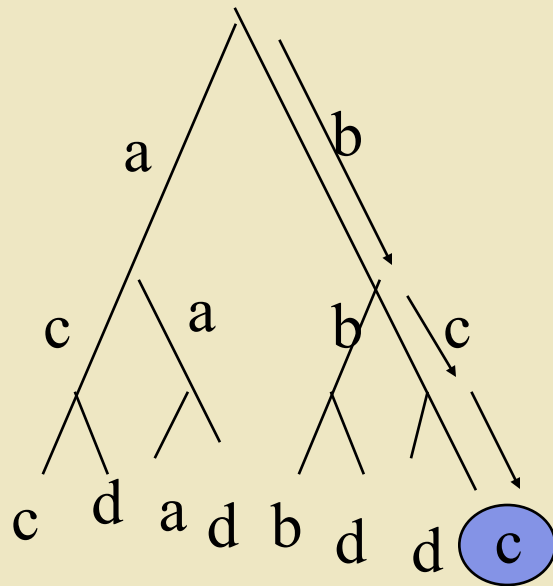
Binary protocol



35% of agents have preferences $c > d > b > a$

35% of agents have preferences $a > c > d > b$

32 % of agents have preferences $b > a > c > d$



35% of agents have preferences $c > d > b > a$

35% of agents have preferences $a > c > d > b$

32 % of agents have preferences $b > a > c > d$



Borda Protocol

- ◆ Borda counts assigns an alternative $|O|$ points whenever it is highest in some agent's preference, $|O|-1$ whenever it is the second and so on.
- ◆ However, Bordas' protocol can also lead to paradoxical results via irrelevant alternatives (see table below)



agent	Preferences
1	$a > b > c > d$
2	$b > c > d > a$
3	$c > d > a > b$
4	$a > b > c > d$
5	$b > c > d > a$
6	$c > d > a > b$
7	$a > b > c > d$
Borda count	C wins with 20, b has 19, a has 18, d loses with 13
Borda count with d removed	A wins with 15, b has 14, c loses 13



Strategic (Insincere) Voter

- ◆ In reality, agents usually have to reveal or declare their preferences.
- ◆ Mechanism design explores interaction mechanisms among rational agents. The goal is to generate protocols such that when agents use them according to certain stability solution concept, then desirable social outcomes follow.



Revelation Principle

A theorem

- ◆ Suppose some protocol (which may include multiple steps) implements social choice function $f(\cdot)$ in Nash (or dominant strategy) equilibrium (where the agents' strategies are not necessarily truthful). Then $f(\cdot)$ is implementable in Nash (or dominant strategy, respectively) equilibrium via a single step protocol where the agents reveal their entire types truthfully.



Gibbard-Satterthwaite Impossibility Theorem

- ◆ Let each agent's type θ_i consists of a preference order $>_i$ on O . Let there be no restrictions on $>_i$, i.e. each agent may rank the outcomes O in any order. Let $|O| \geq 3$. Now, **if the social choice function $f(\cdot)$ is truthfully implementable in a dominant strategy equilibrium then $f(\cdot)$ is dictatorial**, i.e. there is some agent i who gets (one of) his most preferred outcomes chosen no matter what types the others reveal.
- ◆ (Non-manipulable protocols are dictatorial)



Circumventing G-S Impossibility theorem

- ◆ Restricted preference and Groves-Clarke Tax Mechanism
- ◆ Decisive vote is pulled out randomly from a hat (avoid dictatorship)
- ◆ Make counter speculation costly



Quasi-linear preference

- ◆ Utility function is quasi-linear

$$o = (g, \pi_1, \pi_2, \dots, \pi_{|A|})$$

π_i is the amount of numeraire (money) that agent i receives in the outcome

g encodes the other features of the outcome

- ◆ If an agent's utility function is $u_i(o) = v_i(g) + \pi_i$
it is called quasi linear
- ◆ For example, voting for building a joint pool, say, $g = 1$,
and $g = 0$ otherwise



Building Pool example

- ◆ Each agent get gross benefit $v_i^{\text{gross}}(g)$ and the pool cost P
- ◆ The cost is divided equally among agents,

$$\text{namely, } \pi_i = P/|A|$$

Each agent's net benefit is $v_i(g) = v_i^{\text{gross}}(g) - P/|A|$

- ◆ Quasilinear implies: no agent should care how others divide payoffs among themselves and an agent's valuation $v_i^{\text{gross}}(g)$ of the pool should not depend on the amount of money that the agent will have.
- ◆ To vote for the pool imposes an externality on the others because the others have to pay as well. If only pro-pool voters have to pay then there will be incentive for them to vote for no pool.
- ◆ The solution is to make the agents precisely **internalize the externality by imposing a tax on those agents whose vote changes the outcome.**
- ◆ **The size of an agent's tax is exactly how much his vote lowers the others' utility.**



Groves-Clarke Tax Algorithm

- ◆ Every agent $i \in A$ reveals his valuation $v_i(g)$ for every possible g
- ◆ The social choice is $g^* = \operatorname{argmax}_g \sum_i v_i(g)$
- ◆ The social choice without agent i ,
$$g^{*'} = \operatorname{argmax}_g \sum_{k \neq i} v_k(g)$$
- ◆ Every agent is levied a tax:
$$\text{tax}_i = \sum_{j \neq i} (v_j(g^*) - v_j(g^{*'})).$$



Theorem

- ◆ If each agent has quasi linear preference then under Clarke Tax Algorithm, each agent's dominant strategy is to reveal his true preferences, i.e., $\underline{v}_i(g) = v_i(g)$ for $\forall g$



Good thing for Clarke Tax

- ◆ The pool example, the utility each agent i becomes

$$u_i(o) = v_i(1) - P/|A| - \text{tax}_i$$

$$u_i(o) = v_i(0)$$

- ◆ The mechanism leads to the socially most preferred g to be chosen, and no counterspeculation is needed. (truth-telling is the dominant strategy)



Problems of Clarke Tax

- ◆ Too much tax is collected. The tax cannot re-distribute back to agents.
 - So the mechanism is not Pareto-efficiency.
- ◆ The mechanism is not coalition proof. Some coalition of voters might coordinate their insincere preference revelations and achieve higher utilities.

Collusion example

			No collusion				Agent 1 and 2 collude			
i	$V_i^{\text{gross}}(1)$	$V_i(1)$	$V_i(1)$	g^*	tax_i	u_i	$V_i(1)$	g^*	tax_i	u_i
1	5,000	2,000	2000	1	1500	500	2500	1	1000	1000
2	4,000	1,000	1000		500	500	1500		0	1000
3	500	-2,500	-2500		0	-2500	-2500		0	-2500

$P=9000, v_i(0)=0$



Auctions

- ◆ Private value
- ◆ Common value
- ◆ Correlated value



Auction protocols

- ◆ English (first price; open-cry)
- ◆ Japanese (ascending with drop out)
- ◆ First Price sealed bid
- ◆ Dutch auction (first price; descending)
- ◆ Vickrey auction (Second price; sealed bid)



English auction

- ◆ A bidder's dominant strategy is to bid a small amount higher than the current bid price until his private value is reached



Japanese Auction

- ◆ Similar to English ascending auction in which the price rises continuously with bidders choosing when to drop out. When all but one bidder drops out, the good is allocated to the remaining bidder at the price at which the second-to-last bidder dropped out.



First Price Sealed bid auction

- ◆ The highest bidder wins the item and pays the amount of his bid.
- ◆ An agent's best strategy is to bid less than his true valuation, but how much depends on what the others bid.
- ◆ In a private value auction where the valuation v_i for each agent i is drawn independently from a uniform distribution between 0 and v there is a Nash equilibrium where every agent i bid $(\frac{|A|-1}{|A|})v_i$



Dutch (descending) Auction

- ◆ The seller continuously lowers the price until one of the bidders takes the item at the current price.
- ◆ The Dutch auction is strategically equivalent to the first price sealed bid auction.
- ◆ Dutch auctions are efficient in terms of real time because the auctioneer can decrease the price at a brisk pace.



Chinese Auction

- ◆ In Netherlands, Dutch auction is actually known as a "**Chinese auction**".
- ◆ “Chinese Auction
- ◆ Each agent buy tickets for the prize
- ◆ The tickets of all agents are put in a hat
- ◆ A ticket is drawn from the hat which reveals the winner
- ◆ Can repeat many times for many prizees



Vickrey auction (second price, sealed bid)

- ◆ Each bidder submits one bit without knowing the others' bids.
- ◆ The highest bidder wins but at the price of the second highest bid.
- ◆ Theorem: A bidder's dominant strategy is a private value Vickrey auction is to bid his true valuation.



Properties of the auctions

- ◆ The 4 auctions are Pareto efficient because they allocate the auctioned item to the bidder who value them the most.
- ◆ But Vickrey and English auctions (with dominant strategies) are more efficient in the sense that they prevent counterspeculating the other agents.



Revenue equivalence

- ◆ Which auction auctioneer prefer?
 - First price or second price auctions?
- ◆ It turns out that two effects are equally.
- ◆ **Revenue equivalence theorem**
 - All 4 auction protocols produce the same expected revenue to the auctioneer in private value auctions where the values are independent distributed and bidders are **risk neutral. (Why?)**



Which auctions are beneficial to the auctioneer when bidders are risk averse?

- ◆ Among **risk averse** bidders, the Dutch and the first price sealed-bid protocols give higher expected revenue to the auctioneer than the Vickrey or English auction protocol. (Why?)



Winner's curse

- ◆ In a common value auction, the winner might tend to pay more on average in the actual price in the market.



Bidder Collusion Problems

- ◆ All four auctions are not collusion-proof.
- ◆ English auction and the Vickrey auction actually self-enforce bidder to collude.
- ◆ First price sealed bid and Dutch auction are preferred to deter collusion.
- ◆ If Smith has value 20, every other bidder has value 18. The bidders can collude by Smith bidding 6 while others 5.



Collusion in auction

- ◆ In English auction, this is self-enforcing because if one of the other agents exceeds 5, Smith will observe and will be willing to go up to 20.
- ◆ In Vickrey auction, Smith can just bid 20 , because he will get the item for 5 anyway.
- ◆ The bidding of 20 remove the incentives for other to break the coalition agreement.
- ◆ On the other hand,in a sealed bid auction if Smith bid below 18, the other agents will have incentive to bid.



Collusion in auction

- ◆ In order to collude, first sealed bid, Vickrey auction and Dutch auction, all the bidders must identify themselves before bidding, otherwise other bidders will win the auction.
- ◆ English auction is not necessary.
- ◆ A hiding-identities English auction can be designed to deter collusion.



Lying Auctioneer

- ◆ In Vickrey auction, the auctioneer can overstate the second highest bid to the highest bidder unless the bidder can verify it.
- ◆ To solve the problem, cryptographic signature could be used by the bidders so that the auctioneer could actually present the second best bid to the winning bidder and would not be able to alter it.
- ◆ The other three auction protocols, do not suffer the auctioneer cheating problem, because the winner pays the highest bidding price.



Lying Auctioneer

- ◆ In English auction, the auctioneer can “shill” that bid in the auction in order to make the real bidders increase their valuations of the item. [No possible in sealed bid]



Bidders Lying in Non-Private-Value Auction

- ◆ Most auctions are not pure private value auction, an agent's evaluation of a good depends at least in part on the other agents' valuations of the good.
- ◆ Common value (and correlated value) auctions suffer from the winner's curse. If an agent bids its valuation and wins the auction, it will know that its valuation was too high because the other agents bid less. So agents should bid less than their true valuation.
- ◆ Even in Vickrey auction, truthful bidding fails.



Undesirable private Information Revelation

- ◆ Reveal the true valuation might not be desirable for the bidders.
- ◆ A contractor might not want their true evaluation of cost to be revealed to their subcontractors.
- ◆ Vickrey auctions is not suitable in situation where the information of true evaluation is sensitive.

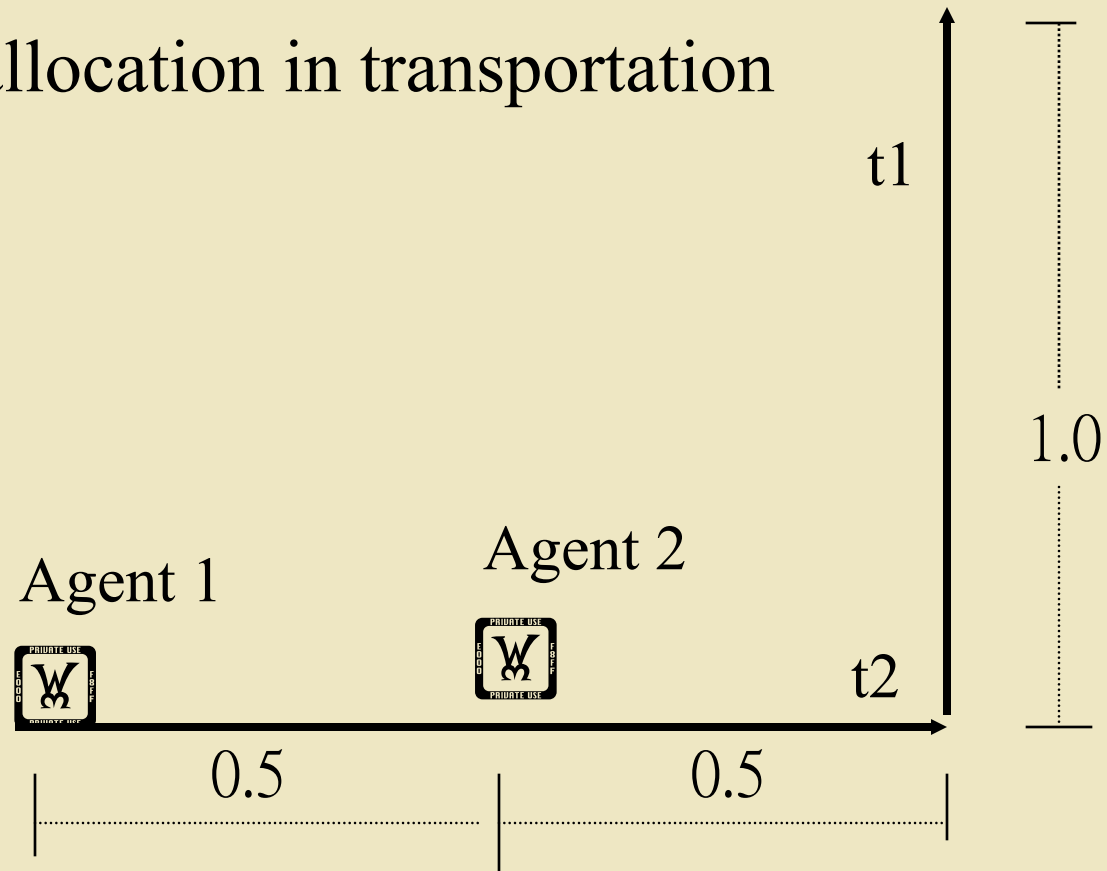


Roles of Computation in Auctions

- ◆ The computational complex lookahead that arises when auctioning interrelated items one at a time
- ◆ The implications of costly local marginal cost (valuation) computation or information gathering in a single-shot auction

Inefficient Allocation and Lying in Interrelated Auctions

- ◆ Task allocation in transportation





Truthful bidding lead to sub-optimal solution

- ◆ $C1(\{t1\})=2.0, c1(\{t2\})=1.0, c1(\{t1,t2\})=2.0$
- ◆ $C2(\{t1\})=1.5, c2(\{t2\})=1.5, c2(\{t1,t2\})=2.5$
- ◆ If the allocation is bid independently, and agents bid truthfully
- ◆ T1 will be allocated to agent2 and T2 will be allocated to agent1 $1.5+1.0=2.5$
- ◆ The result is sub-optimal since if t1, t2 are allocated to agent1 will cost only 2.0



Bidding Order is Considered

- ◆ However, if bidding order is t_1 and then t_2
- ◆ Then, after agent2 bid the t_1 with cost 1.5
- ◆ The additional cost for agent2 to bid t_2 becomes $c_2(\{t_1, t_2\}) = 2.5 - 1.5 = 1.0$

So agent2 will bid t_2 for 1.0 instead of 1.5 before

- ◆ In this case t_2 will be assigned to either agent1 or agent2 equally likely



Look-ahead is considered

- ◆ If agent1 has t_1 then he will bid t_2
 $c_1(\{t_1, t_2\}) - c_1(\{t_1\}) = 2.0 - 2.0 = 0$
Otherwise $c_1(\{t_2\}) = 1.0$
- ◆ If agent2 has t_1 then he will bid t_2
 $c_2(\{t_1, t_2\}) - c_2(\{t_1\}) = 2.5 - 1.5 = 1$
Otherwise $c_2(\{t_2\}) = 1.5$

- 
- ◆ So if agent 1 has t_1 , it will win t_2 at the price 1.5 and get a payoff $1.5 - 0 = 1.5$ in the second auction while agent 2 gets zero.
 - ◆ On the other hand, if agent 2 has t_1 , the bids are equal, and both agent gets a zero payoff in the second auction irrespective of who t_2 gets allocated to.

- 
- ◆ Therefore it is known that getting t_1 in the first auction is worth an extra 1.5 for agent 1 while nothing extra for agent 2.
 - ◆ So in the auction 1, agent 1's dominant strategy is to bid $c_1(\{t_1\}) - 1.5 = 2 - 1.5 = 0.5$
 - ◆ This is lower than agent 2's bid $c_2(\{t_1\}) - 0 = 1.5 - 0 = 1.5$
 - ◆ So agent 1 gets t_1 in the first auction and gets t_2 also in the second auction as discussed. And the globally optimal allocation is reached.



Interrelated items and Vickrey auction

- ◆ Lookahead is a key feature in auctions if multiple interrelated items.
- ◆ In human auctions, interrelated items
 - Allow pooling all interrelated items under one entire bid.
 - Allow agents to backtrack from commitments by paying penalties



Counter-speculation When Computing One's Valuation

- ◆ Even Vickrey auction protocol fails to avoid counter-speculation.
- ◆ Suppose an agent is uncertain about its own valuation of the auction item and can pay to remove the uncertainty.
 - The cost of computation of the value of the item



Incentive to Counter Speculation

- ◆ Theorem:

In a single-shot private value Vickrey auction with uncertainty about an agent's own valuation, a risk neutral agent's best (deliberation or information gathering) action can depend on the other agents. It follows that (if counter speculation is cheap enough) it is worth counter speculation.



Proof by example

- ◆ Agent 1's valuation v_1 be uniformly distributed between 0 and 1, agent 1 doesn't know its value exactly.
- ◆ Agent 2's valuation v_2 be common knowledge $0 \leq v_2 \leq \frac{1}{2}$ which implies $E[v_1] > v_2$
- ◆ No matter agent 1 bids, agent 2 will bid v_2 (Vickrey auction true bidding strategy is the dominant strategy)

agent 1 will bid $E[v_1] = \frac{1}{2}$

- 
- ◆ So $E[\pi_{\text{no-info}}] = 1/2 - v_2$
 - ◆ If agent 1 choose to pay cost c for the exact information, agent 1 gets the item iff $v_1 \geq v_2$
 - ◆ So the expected payoff of getting information is
$$\begin{aligned} E[\pi_{\text{info}}] &= \int_0^{v_2} (-c) dv_1 + \int_{v_2}^1 (v_1 - v_2 - c) dv_1 \\ &= -cv_2 + (1/2v_1^2 - v_2v_1 - cv_1)|_{v_2}^1 \\ &= -cv_2 + (1/2 - v_2 - c) - (1/2v_2^2 - v_2^2 - cv_2) \\ &= 1/2v_2^2 + 1/2 - v_2 - c \end{aligned}$$

- 
- ◆ Agent 1 should choose to buy the information iff $E[\pi_{\text{info}}] \geq E[\pi_{\text{no-info}}]$
 - $\Leftrightarrow 1/2v_2^2 + 1/2 - v_2 - c \geq 1/2 - v_2$
 - $\Leftrightarrow 1/2v_2^2 \geq c$
 - $\Leftrightarrow v_2 \geq \sqrt{2c}$

Agent 1's best choice depends on agent 2's valuation v_2



Combinatorial Auction (1)

- ◆ In some cases, a buyer's value for the goods for auction is a complex interaction of the type and number of goods he receives (known as a "bundle").
- ◆ For instance, if bicycle wheels and bicycle frames are sold separately in an auction, a bidder may value a bundle consisting of a single wheel or a single frame at \$50, but may value the bundle of two wheels and one frame at \$200. If forced to purchase each component of a bundle in a separate auction, the bidder faces a dilemma: bidding enough to win the components of the bundle that are sold first may result in a financial loss if he fails to win the components that are sold later, but failing to purchase the components that are sold first ensures that he will not win the bundle.
- ◆ This dilemma can be overcome by selling all goods simultaneously and allowing buyers to submit bids on combinations of goods.



Combinatorial auction (2)

- ◆ Such combinatorial bids may offer to pay a certain amount if all units of a buyer-specified bundle are awarded, but nothing otherwise.
- ◆ They may also offer to purchase one bundle of goods or another, but not both.
- ◆ Sorting out which buyers win which bundles (and sometimes the amount they must pay for them) is usually computationally complex.
- ◆ This complexity is overcome by feeding the bids into an optimization algorithm (such as a linear programming problem).



Continuous double auctions

- ◆ Agent bidders could be a seller or a buyer during the auction period.
- ◆ Stock market is a CDA.



Reverse auctions

- ◆ While traditional auctions involve a single seller and many buyers, a reverse auction generally involves many sellers and one buyer. For example, procurement auctions are used to obtain competitive bids to provide goods or services. Theoretically, regular auctions and reverse auctions are equivalent.



Trading Agent Competition

- ◆ <http://www.sics.se/tac/page.php?id=1>
- ◆ The Trading Agent Competition (TAC) is an international forum designed to promote and encourage high quality research into the trading agent problem. Michael Wellman led the team that organized the first years competitions. based on the "travel agent scenario" (called TAC Classic). From TAC 2002, SICS organized the competition together with the TAC community.
- ◆ Since 2003 the competition also has a supply chain scenario created by Carnegie Mellon University and SICS (called TAC SCM).
- ◆ Initiated by Mike Wellman, U of Michigan



Further reading about auctions

- ◆ Klemperer, Paul (2004). *Auctions: Theory and Practice*. ISBN 0-691-11925-2.
<http://www.nuff.ox.ac.uk/users/klemperer/VirtualBook/VirtualBookCoverSheet.asp>
- ◆ Paul Milgrom, Putting auction theory to work: The Simultaneous Ascending Auction, 1998.



Bargaining

- ◆ When agents have conflicts they make mutually beneficial agreement
 - Monopolist gets all gains
 - Perfect competition can make no profit
 - Bargaining fit in between
- ◆ Two bargaining fields:
 - Axiomatic Bargaining theory
 - Strategic Bargaining Theory



Axiomatic Bargaining Theory

- ◆ Nash Bargaining Solution is the solution concept
- ◆ o_{fallback} : When no agreement is reached
- ◆ Assume utility function $u_i: O \rightarrow \mathbb{R}$ for each agent $i \in [1, 2]$
- ◆ The set of feasibility utility vector $\{(u_1(o), u_2(o)) | o \in O\}$ is convex
- ◆ Individual rationality: $u(o) \geq u(o_{\text{fallback}})$



Axioms for Nash Bargaining Solution

- ◆ $u^*=(u_1(o^*),u_2(o^*))$ must satisfy
- ◆ **Invariance**: The numeric utility functions really only represent ordinal preferences among outcomes, the actual cardinality of the utilities do not matter. Therefore it is possible to transform the utility functions: $u^*(f(o),f(o_{\text{fallback}}))=f(u^*(o,o_{\text{fallback}}))$
- ◆ **Anonymity(symmetry)**: switching labels on agents does not affect the result
- ◆ **Independence of irrelevant alternatives**: If some outcomes o are removed but o^* is not then o^* still remains the solution
- ◆ **Pareto efficiency**: It is not feasible to find an outcome that gives both agent higher utility than u^*



Nash Bargaining Solution Theorem

- ◆ The unique solution that satisfies four axioms is

$$o^* = \operatorname{argmax}[u_1(o) - u_1(o_{\text{fallback}})][u_2(o) - u_2(o_{\text{fallback}})]$$

- ◆ 2-agent Nash bargaining solution is a special case of Shapley value



Strategic Bargaining Theory

- ◆ Does not postulate fairness axioms
- ◆ Explain rational expected utility maximizing behaviors of agents
- ◆ Strategic bargaining theory usually analyses sequential bargaining where agents alternate in making offer to each other in a pre-specified order.



Strategy Bargaining

Consider how to split a dollar:

- ◆ In **finite number of offers** and **no time discount**
 - The unique payoffs of subgame perfect Nash equilibria are such that the last offerer will get the whole dollar (minus ϵ)
 - because the other agent is better off accepting ϵ than by rejecting and receiving nothing.
 - Even ϵ is zero the other agent still accept it



Subgame perfect equilibrium

- ◆ In game theory, a **subgame perfect equilibrium** is a refinement of a Nash equilibrium used in dynamic games. A strategy profile is a subgame perfect equilibrium if it represents a Nash equilibrium of every subgame of the original game.
- ◆ More informally, this means that if (1) the players played any smaller game that consisted of only one part of the larger game and (2) their behavior represents a Nash equilibrium of that smaller game, then their behavior is a subgame perfect equilibrium of the larger game.



Strategy Bargaining: Finite with time discount

- ◆ A time discount factor δ
 - The dollar is worth 1 at round 1
 - It is worth δ at round 2
 - It is worth δ^2 at round 3,...
- ◆ With the time discount, a subgame perfect Nash equilibrium of a **finite game of length T** can be solved starting from the end.



Table

◆ Round	1's share	2's share	total value	offerer
...	...	...	...	...
T-3	0.819	0.181	0.9^{T-4}	2
T-2	0.91	0.09	0.9^{T-3}	1
T-1	0.9	0.1	0.9^{T-2}	2
T	1	0	0.9^{T-1}	1



Infinite non-discount bargaining

- ◆ The solution concept is powerless because any split of the dollar can be supported in subgame perfect Nash equilibrium just as in the single-shot case.
- ◆ However, **with discounting**, the infinite game can be solved.



Rubinstein Bargaining Solution

- ◆ In a discounted **infinite round** setting, the subgame perfect Nash equilibrium outcome is unique.
- ◆ Agent 1 gets
$$(1-\delta_2)/(1-\delta_1\delta_2),$$
Agent 2 gets 1 minus this
- ◆ **Agreement is reached at the first round**



Table (infinite game with different discount factors)

◆ Round	1's Share	2's Share	Offerer
...			...
t-2	$1 - \delta_2(1 - \delta_1\pi_1)$		1
t-1		$1 - \delta_1\pi_1$	2
t	π_1		1
...	...	...	...



Proof

- ◆ Let π_1 be the maximum undiscounted share that 1 can get in any subgame perfect Nash equilibrium on his turn to offer.
- ◆ $\pi_1 = 1 - \delta_2(1 - \delta_1\pi_1) \iff$
$$\pi_1 = (1 - \delta_2) / (1 - \delta_1\delta_2)$$
- ◆ Proof with the same logic for minimum undiscounted share,...
- ◆ Minimum is equal to maximum



Sequential Bargaining with Fixed Cost (not discount)

- ◆ If both have the same cost, any split of dollar can be supported by the subgame perfect Nash equilibrium
- ◆ If c_1 is less than c_2 , the agent 1 gets the entire dollar
- ◆ If c_1 is greater than c_2 , then the agent1 gets the payoff c_2 and agent 2 gets the rest.



Computation in Bargaining

- ◆ Agents in bargaining conduct two kinds of search:
- ◆ Deliberative search: generate alternatives, evaluate them, counterspeculate, does lookahead, etc.
- ◆ Inter agent committal search: make agreements with each other regarding the solution.



General Equilibrium Market Mechanism

- ◆ General equilibrium theory: a general market framework for resource allocation in microeconomic.
- ◆ Market has n Commodity goods: g
- ◆ Corresponding prices: $p = [p_1, \dots, p_n]$
- ◆ Consumers I and producers J
- ◆ Consumption bundles for agent i : $x_i = [x_{i1}, \dots, x_{in}]$
- ◆ Endowment for agent i , $e_i = [e_{i1}, \dots, e_{in}]$
- ◆ Production vector $y_j = [y_{j1}, \dots, y_{jn}]$
- ◆ θ_{ij} the fraction of producer j that consumer i owns



General Equilibrium Market Mechanisms

- ◆ We say that (p^*, x^*, y^*) is a general (Walrasian) equilibrium if

- ◆ I. Market clearance:

$$\sum_i x_i^* = \sum_i e_i + \sum_j y_j^*$$

- ◆ II. Each consumer i maximize its preferences given the prices

$$x_i^* = \operatorname{argmax}_{x_i \text{ s.t. } p^* x_i \leq p^* e_i + \sum_j \theta_{ij} p^* y_j^*} u_i(x_i)$$

- ◆ III. Each producer j maximizes its profits given the prices

- ◆ $y_j^* = \operatorname{argmax}_{y_j \in Y_j} p^* y_j$



Properties of general equilibrium

- ◆ **Theorem Pareto efficiency:**

Each general equilibrium is Pareto efficient, I.e. no agent can be made better off without making other agent worse off

- ◆ **Theorem Coalitional stability:**

Each general equilibrium with no producers is stable in the sense that no subgroup of consumers can increase their utilities by pulling out of the equilibrium and forming their own market



- ◆ When producers are present, the situation becomes more complex, consumers might collude...
- ◆ Unfortunately, in some domains no general equilibrium exists, for example, some producer produce infinite amount of good..., however, the sufficient conditions for existence are known...



Existence of general equilibrium

- ◆ Let **the production possibility sets** be closed (include their boundaries), convex (if bundles y and y' are producible, then so is $\alpha y + (1 - \alpha)y'$, $\forall \alpha \in [0, 1]$) and bounded above (no infinite amount of good can be produced).
- ◆ Let the **consumer's preferences** be continuous, strictly convex (if consumers prefers y to y' and y' to y'' then he prefers $\alpha y + (1 - \alpha)y'$ to y'' $\forall \alpha \in [0, 1]$) and strongly monotone (each consumer strictly prefers more to less of each commodity).
- ◆ If a society-wide bundle is producible where the amount of each commodity is positive (**positive endowments trivially imply this**), a general equilibrium exists.



Violation of general equilibrium assumptions

- ◆ **Continuity of Consumer's preferences** is violated in bandwidth allocation if agent's welfare jumps as the threshold for being able to participate in a video conference is reached.
- ◆ **Consumer's preferences are not convex**, if the consumer starts to prefer a good more as he gets more of that good. [Drug, Web surfing]



Theorem: Uniqueness under gross substitute

- ◆ A general equilibrium is unique if society-wide demand for each good is **non-decreasing** in the prices of the other goods
- ◆ E.g. Prices of meat increases, the demand for potatoes increases.
- ◆ However, prices of bread increases, the demand of butter decreases, the theorem does not satisfy...



Externality

- ◆ The general equilibrium does not account for externalities.
- ◆ **Consumption externalities**, one agent's consumption affects another agent's utility.
- ◆ **Production externalities**, one agent's production possibilities set is directly affected by another agent's actions.
- ◆ This is a field of **computational ecosystems**.



Distributed search for a general equilibrium

- ◆ The distributed price Tatonnement algorithm: a steepest descent search method
 - Algorithm for the price adjustor
 - Algorithm for consumer I
 - Algorithm for producer J



Algorithm for the price adjustor

- ◆ $P_g=1$ for all $g \in [1..n]$
- ◆ Set λ_g to a positive number for all $g \in [1..n]$ *
- ◆ Repeat
 - Broadcast p to consumers and producers
 - Receive a production plan y_j from each producer j
 - Broadcast the producer plans y_j to consumers
 - Receive a consumption plan x_i from each consumer i
 - For $g=1$ to n *

$$P_g = P_g + \lambda_g (\sum_i (x_{ig} - e_{ig}) - \sum_j y_{jg}) \text{ until}$$
$$| \sum_i (x_{ig} - e_{ig}) - \sum_j y_{jg} | < \epsilon \text{ for all } g \in [1..n]$$

Inform consumers and producers the equilibrium has been reached



Algorithm for consumer i

Repeat

Receive p from the adjustor

Receive a production plan y_j for each j from the adjustor

Announce to the adjustor a consumption plan $x_i \in \mathbb{R}_+^n$ that maximizes $u_i(x_i)$ given the budget constraint

$$p \cdot x_i \leq p \cdot e_i + \sum_j \theta_{ij} p \cdot y_j$$

until informed that an equilibrium has been reached

Exchange the consume



Algorithm for producer j

Repeat

Receive p from price adjustor

Announce to the adjustor a production plan $y_j \in Y_j$ that
maximize $p \bullet y_j$

until informed that an equilibrium has been reached
exchange and produce



Convergence Theorem

- ◆ The price Tatonnement algorithm converges to a general equilibrium if
$$p^* \cdot (\sum_i (x_i(p) - e_i) - \sum_j y_j(p)) > 0$$
for all p not proportional to an equilibrium vector p^*

Newton Method

$$X_{n+1} = x_n - f(x_n)/f'(x_n)$$

- ◆ $dp_g/dt = \lambda_g (\sum_i (x_{ig}(p) - e_{ig}) - \sum_j y_{jg}(p))$
- ◆ If multiplier λ_g is sufficiently small, the algorithm often converges. Large λ_g will overshoot the equilibrium while small λ_g will cause slow convergence.
- ◆ $dp_g/dt = \lambda_g [J(p)]^{-1} (\sum_i (x_{ig}(p) - e_{ig}) - \sum_j y_{jg}(p))$
- ◆ $J(p) = \begin{pmatrix} \partial(\sum_i (x_{i1}(p) - e_{i1}) - \sum_j y_{j1}(p)) / \partial p_1 & \dots & \partial(\sum_i (x_{i1}(p) - e_{i1}) - \sum_j y_{j1}(p)) / \partial p_n \\ \vdots & & \vdots \\ \partial(\sum_i (x_{in}(p) - e_{in}) - \sum_j y_{jn}(p)) / \partial p_1 & \dots & \partial(\sum_i (x_{in}(p) - e_{in}) - \sum_j y_{jn}(p)) / \partial p_n \end{pmatrix}$



Centralized optimization by newton method

- ◆ Newton Method requires fewer iterations than steepest descent, but each iteration is computational more intensive for the price adjustor and requires the computation and communication of the derivation information from consumers and producers.
- ◆ One can assume that all the functions are submitted to the price adjustor, but this violates the assumption that the market is distributed.



WALRAS simulation

- ◆ WALRAS uses asynchronous declaration by the agents, agents might only change their plans regarding a subset of goods at a time
- ◆ agents might take arbitrary turns in making new declaration
- ◆ Under certain conditions, this process converges to a general equilibrium



Quantity-based mechanism

- ◆ In addition to price-based mechanism, Quantity-based, commodity-based and resource-based methods for reaching a general equilibrium exist.
- ◆ Quantity-based maintain a feasible solution where market clears at every iteration, this makes it good for anytime algorithm.



Computation cost for demand and supply functions

- ◆ Most market-based search only discuss the complexity of finding an equilibrium once the agent's supply and demand function are known.
- ◆ However, it may be computational complex for each agent to generate its optimal supply and demand function given current prices.
- ◆ Each agent has to go through a local deliberation at every iteration of the market protocol because price change affects the optimal plan for each agent.



Case A: Speculating Strategies in Equilibrium Markets

- ◆ The goal of a self-interest consumer is to find the consumption bundle that maximizes its utility.
- ◆ To find the optimal bundle when acting in an equilibrium market, the consumer must speculate how other agents respond to the prices.
- ◆ This is because its demand function decision affect the prices which affect the demand and supply decision of others, which again affect the prices that the consumer faces.



Case A: Speculating Strategies in Equilibrium Markets

- ◆ Using the model of others, the consumer computes its optimal demand decisions.
- ◆ A solution of a speculating consumer is following:

- ◆ $\max u_s(x_s(p))$ s.t.

$$x_{sg}(p) \geq 0 \quad [\text{consumer does not produce}]$$

$$x_{sg}(p) = e_{sg} - (\sum_{i \in \text{Consumers} - \{s\}} (x_{ig} - e_{ig}) - \sum_j y_{jg})$$

[Supply meets demand]

$$p * (x_s - e_s) \leq \sum_j \theta_{sj} y_j(p) \quad [\text{budget constraint}]$$



Case B: Speculating Producer

- ◆ The goal of a self-interested producer is to find the production vector that maximizes its profits.
- ◆ A solution to the maximization problem gives the highest profit that a speculating producer s can obtain is following:
- ◆ $\max_p p^* y_s(p)$ s.t.
 $y_s(p) \in Y_s$ [feasible production plan]
 $y_{sg} = \sum_i (x_{ig} - e_{ig}) - \sum_{j \in \text{Producers} - \{s\}} y_{jg}$
[Supply meets demand]



Reaching Equilibrium under speculation: Driving the Market

- ◆ P^{**} is the solution to the optimization under which a consumer or a producer is speculating.
- ◆ P^{**} is not Pareto efficient in general.
- ◆ A speculating agent's best strategy is to declare demand plans $x_s(p)$ (or production plan $y_s(p)$) such that the market clears at the desired prices p^{**} (an equilibrium exists) and the market process will find it.

- 
- ◆ The market clears at p^{**} if for each good g ,

$$x_{sg}(p^{**}) = e_{sg} - (\sum_{i \in \text{Consumers} - \{s\}} (x_{ig}(p^{**}) - e_{ig}) - \sum_j y_{jg}(p^{**}))$$

$$y_{sg}(p^{**}) = \sum_i (x_{ig}(p^{**}) - e_{ig}) - \sum_{j \in \text{Producers} - \{s\}} y_{jg}(p^{**})$$

[5.3]

- ◆ What remains to be analyzed is whether the particular market algorithm finds the equilibrium even if the speculator acts strategically.



Conditions for finding market equilibrium

- ◆ Many standard market algorithms e.g. price tatonnement, Newtonian price tatonnement, and WALRAS are guaranteed to find the equilibrium if
- ◆ $\partial(\sum_i (x_{ig}(p) - e_{ig}) - \sum_j y_{jg}(p)) / \partial p_g < 0$
[Society-wide demands decreases as price increases]
- ◆ $\partial(\sum_i (x_{ig}(p) - e_{ig}) - \sum_j y_{jg}(p)) / \partial p_h \geq 0$ for goods $g \neq h$
[goods are gross substitutes]



Conditions for converging to the unique optimal solution for speculating agents

- ◆ Let us assume that the above two conditions hold in the market when speculators were not present, now if
 - 1) the speculating agent uses the strategy that satisfies the equations (5.3)
 - 2) $\partial(x_{sg}(p) - e_{sg}) / \partial p_g \leq 0$ if s is a consumer, $\partial y_{sg}(p) / \partial p_g \geq 0$ if s is a producer
 - 3) For goods $g \neq h$ $\partial(x_{sg}(p) - e_{sg}) / \partial p_h \geq 0$ if s is a consumer, $\partial(x_{sg}(p) - e_{sg}) / \partial p_h \leq 0$
- ◆ The market is guaranteed to the unique equilibrium prices p^{**} that maximize the speculator's gain.



Simple generic Strategies

- ◆ The speculators will be able to drive the market to the equilibrium where his maximal gain from speculation materializes.

- ◆ A linear strategy for a consumer:

$$x_s(p) = -e_s + p^{**} - p - \left(\sum_{i \in \text{Consumers} - \{s\}} (x_{ig}(p^{**}) - e_{ig}) - \sum_j y_{jg}(p^{**}) \right)$$

- ◆ A constant strategy

$$x_s(p) = -e_s - \left(\sum_{i \in \text{Consumers} - \{s\}} (x_{ig}(p^{**}) - e_{ig}) - \sum_j y_{jg}(p^{**}) \right)$$

- ◆ Corresponding strategy for a producer:

$$y_{sg}(p) = p - p^{**} + \sum_i (x_{ig}(P^{**}) - e_{ig}) - \sum_{j \in \text{Producers} - \{s\}} y_{jg}(P^{**})$$

And

$$y_{sg}(p) = \sum_i (x_{ig}(P^{**}) - e_{ig}) - \sum_{j \in \text{Producers} - \{s\}} y_{jg}(P^{**})$$



Convergence Speed

- ◆ In any particular market setting, it maybe that the market converges slower or faster when an agent acts strategically than he acts competitively.



Market Oriented Programming

- ◆ The idea of market-oriented programming is to exploit the institution of markets and our models of them to build computational economies to solve particular problems of distributed resource allocation.



Term project Due June 9th

- ◆ Implement a general market oriented programming using price Tatonnement, Walras algorithms.
- ◆ Assume three consumer agents, two production agents and two goods X_1 and X_2 are sold in the market
- ◆ Assume price adjustor factors λ_1 and λ_2 (for goods 1 and 2) are both 0.1.
- ◆ Initial endowments of consumer agents 1,2,3 for each good x_1 and x_2 are:
$$e_{11} = 3, e_{12} = 5; \quad e_{21} = 4, e_{22} = 2; \quad e_{31} = 2, e_{32} = 6.$$
- ◆ Each consumer agent own shares of production agents (1 & 2) are:
- ◆ $\theta_{11} = 0.8; \theta_{12} = 0.2; \theta_{21} = 0.3; \theta_{22} = 0.7; \theta_{31} = 0.6; \theta_{32} = 0.4$



Utility functions $U_i(x_{ig})$ of agent i for good 1 & 2

$$U_1(x_1) = 2(0.5x_{11}^{0.9} + 0.5x_{12}^{0.9})^{\frac{0.5}{0.9}}$$

$$U_2(x_2) = 3(0.6x_{21}^{0.95} + 0.4x_{22}^{0.95})^{\frac{1.6}{0.95}}$$

$$U_3(x_3) = 5(0.7x_{31}^{0.8} + 0.3x_{32}^{0.8})^{\frac{0.7}{0.8}}$$

- 
- ◆ Assume supply agents has **Cobb-Douglas** Production function on goods x_1 and x_2 :

$$Y_1(x_1, x_2) = k x_1^\alpha x_2^\beta, \text{ where}$$

$$k=2.0; \quad \alpha=0.7; \quad \beta=0.3$$

$$Y_2(x_1, x_2) = k x_1^\alpha x_2^\beta, \text{ where}$$

$$k=3.0; \quad \alpha=0.2; \quad \beta=0.6$$

Production capability for Y_{i1} and Y_{i2} are not greater than 50.

- 
- ◆ You should write a simulation program to run the market and find the general market equilibrium prices and quantities for goods produced.
 - ◆ You should implement the price adjustor, producer and consumer algorithms and then run the simulation.
 - ◆ You could write in any programming language, and execute the simulation runs by changing different initial prices to see if they all converge to the same equilibrium.
 - ◆ You should also change the price adjustor parameters to compare the converging efficiency.

The term project worths 100 points. But if you could implement the price adjuster, producer and consumer in agent programming Jade and execute in Jade platform, it will be a bonus 25 points. Change different utility functions and production functions and report the observation add another 25 points.



Contract Net

- ◆ Contract net protocol is a distributed negotiation in task allocation domain.
- ◆ A new contract net protocol that is not based on cooperative agents only.



Task Allocation Negotiation

- ◆ A task allocation problem is defined by a set of tasks T , a set of agents A , a cost function $c_i: 2^T \rightarrow \mathbb{R} \cup \{\infty\}$ (cost for agent i to handle the task)
- ◆ The initial tasks for agents $\langle T^{\text{init}}_1, \dots, T^{\text{init}}_{|A|} \rangle$ where $\bigcup_{i \in A} T^{\text{init}}_i = T$
- ◆ and $T^{\text{init}}_i \cap T^{\text{init}}_j = \emptyset$ for all $i \neq j$



Individual Rationality

- ◆ A contract is individually rational (IR) to an agent if that agent is better off with the contract than without it.
- ◆ Specifically a contractee q accepts a contract if it gets paid more than its marginal cost of handling the tasks T^{contract} of the contract.

$$MC^{\text{add}}(T^{\text{contract}}|T_q) \equiv c_q(T^{\text{contract}} \cup T_q) - c_q(T_q)$$



A contractor

- ◆ A contractor r is willing to allocate the task T^{contract} from its current task set T_r to the contractee if it has to pay the contractee less than it saves by not handling the tasks T^{contract} itself.

$$MC^{\text{remove}}(T^{\text{contract}}|T_r) \equiv c_r(T_r) - c_r(T_r - T^{\text{contract}})$$



Contract net protocol

- ◆ In the protocol, agents suggest contracts to each other, and make their accepting and rejecting decisions based on these marginal cost calculations.
- ◆ An agent can take on both contractor and contractee roles.
- ◆ It can subcontract out tasks.
- ◆ Assume no knowledge of other agents' cost functions.



Property of the contract net protocol

- ◆ This protocol is hill-climbing in the space of task allocations.
- ◆ The scheme is anytime algorithm: Contracting can be terminated at anytime and the worth of each agent's solution increases monotonically.
- ◆ It follows that social welfare increases monotonically.



Convergence to the globally optimal task allocation

- ◆ Each contract regards only one task, ie, one task is moved from one agent to another against a payment: Original contract (O-contract)
- ◆ When the contract net protocol is equipped with O-contracts only, it may get stuck in a local optimal where no contract is individually rational but the task allocation is not globally optimal.



Theorem I

- ◆ Several contract types: cluster (C) where a set of tasks is atomically contracted from one agent to another, swap (S) contracts, where a pair of agents swap a pair of tasks and multiagent (M) contracts where more than one agents are involved in an atomic exchange of tasks.
- ◆ Each of the four contract types avoids some of the local optimal that the other three do not.
- ◆ For each of the four contract type (O,C,S and M), there exist task allocations where no IR contract with the other three contract types is possible, but an IR contract with the fourth type is.



Theorem II

- ◆ Unfortunately, even if the contracting protocol is equipped with all four of the contract types, the globally optimal task allocation may not be reached via IR contracts
- ◆ There are instances of the task allocation problem where no IR sequence from the initial task allocation to the optimal one exists using O-, C-, S- and M-contracts.



OCSM-contract

- ◆ Without an oracle, contracting may get stuck at local optimal even if an IR sequence exists because the agents may choose some other IR sequence.
- ◆ An OCSM contract is defined by a pair $\langle T, \rho \rangle$ of $|A| \times |A|$ matrices. An element $T_{i,j}$ is the set of tasks that agent i gives to agent j , and an element $\rho_{i,j}$ is the amount that i pays to j .
- ◆ So OCSM contracts allow moving from a task allocation to any other task allocation with a single contract.



Theorem for OCSM contracts

- ◆ Let $|A|$ and $|T|$ be finite. If the contracting protocol allows OCSM-contracts any hill-climbing algorithm (i.e. any sequence of IR contracts) finds the globally optimal task allocations in a finite number of steps.



Theorem for OCSM contracts

- ◆ OCSM-contracts are also necessary: no weaker set of contract types suffices – even if there were an oracle to choose the order in which to apply them.
- ◆ If there is some OCSM-contract that the protocol does not allow, there are instances of the task allocation problem where no IR sequence exists from the initial allocation to the optimal one.



Homework 3

- ◆ Find an example for O,C,S,M respectively that contract net protocol based on IR will stuck at a local optimal.
- ◆ Due May 22th.



Insincere Agents in Task Allocation

- ◆ We assume the contract protocols are based on IR (individual rationality) and agents bid truthfully.
- ◆ However, agents may benefit by exaggerating its marginal cost. On the other hand, lying too much may cause non-IR to be detected and the contract to be rejected.



Contingency Contracts and Leveled Commitment Contracts

- ◆ In traditional negotiation protocol among self-interested agents, once a contract is made, it is binding, ie, neither party can back out.
- ◆ A contract may be profitable to an agent when viewed ex ante, but it is not necessary profitable when viewed ex post.



Leveled Commitment Contracts

- ◆ Leveled commitment contracts allows agents to de-commit from original contract with some penalty.
- ◆ It is shown that even under insincere decommitment, the leveled commitment protocol outperforms the full commitment protocol.



Coalition Formation

- ◆ Nash equilibrium is too weak
- ◆ Strong Nash equilibrium requires no subgroup of agents can get mutual beneficial deviation
- ◆ Coalition-proof Nash equilibrium: no subgroup of agents can get mutual beneficial deviation in a way that deviation itself is stable according to the same criterion
- ◆ Does not guarantee the existence of the solution



Characteristic games

- ◆ Coalition structure
- ◆ Finding a coalition structure that maximize the social welfare

$$CS^* = \operatorname{argmax}_{CS \in \text{partition}} V(CS)$$

$$\text{where } V(CS) = \sum_{S \in CS} v(S)$$

- ◆ However, the # coalition structures is large $\Omega(|A|^{|A|/2})$

- 
- ◆ One search through a subset N of all the partitions of A

$$CS^*_N = \operatorname{argmax}_{CS \in N} V(CS)$$

- ◆ To guarantee a worse case bound from optimal that

$$k \geq V(CS^*)/V(CS^*_N)$$

is finite and as small as possible.

- ◆ Let $n_{\min}$ be the smallest size of N that allows us to establish such a bound



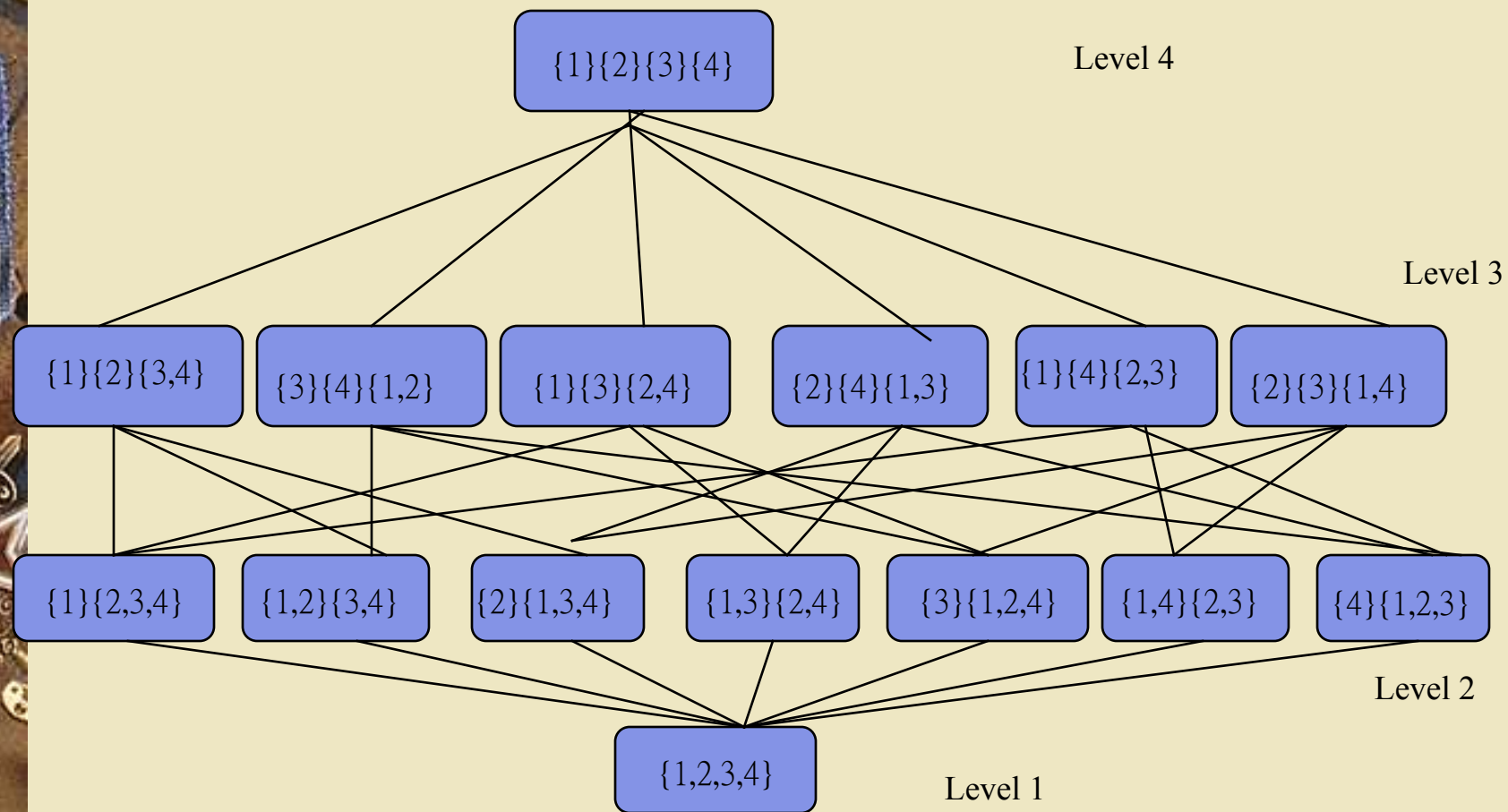
Theorem minimal Search to establish a bound

- ◆ To bound k , it suffices to search the lowest two levels of the coalition structure graph.
- ◆ With this search, the bound $k = |A|$, this bound is tight and the number of nodes searched is $n = 2^{|A|-1}$.
- ◆ No other search algorithm (than the one searches the bottom two levels) can establish a bound k while searching only $n = 2^{|A|-1}$ nodes or fewer.

- 
- ◆ Positive: a worse case bound from optimal can be guaranteed without seeing all CSs.

As $|A| \rightarrow \infty$, $n_{\min}/|\text{Partitions of } A| \rightarrow 0$.

- ◆ Negative: exponentially many coalition structures have to be searched before a bound can be established
 - No algorithm of coalition structure generation algorithm could establish such bound without trying at least $2^{|A|-1}$ coalition structures.





Coalition Structure Search Algorithm

- ◆ Search the bottom two levels of the coalition structure
- ◆ Continue with a breadth-first search from the top of the graph as long as there is time left or until the entire graph has been searched
- ◆ Return the coalition structure that has the highest welfare among those seen so far



- ◆ The algorithm establish a bound in minimal amount of search
- ◆ And then rapidly reduce the bound further if there is time for more search
- ◆ If the domain happens to be **super-additive**, the algorithm finds optimal coalition structure $[V(A \cup B) \geq V(A) + V(B)]$



Theorem lowering the bound with further search

- ◆ After searching level 1 with the algorithm, the bound k is $\lceil |A|/h \rceil$ if $|A| \equiv h-1 \pmod{h}$
- ◆ And $|A| \equiv L \pmod{2}$
- ◆ $h = \lfloor (|A| - L) / 2 \rfloor + 2$



Other Coalition structure generation Algorithms

- ◆ Merging algorithm: (top-down) cannot establish any bound before it has searched the entire graph
- ◆ Splitting algorithm: (bottom-up) establish a bound until the point it reached 2^{a-1} nodes have been searched. But then it reduces the bound slower than the algorithm Tuomas mentioned.



Payoff Division according to Core

- ◆ Core = $\{(\underline{x}, CS) \mid \forall S \subset A, \sum_{i \in S} x_i \geq v_S \text{ and } \sum_{i \in A} x_i = \sum_{S \in CS} v_S\}$
- ◆ Core is a solution concept based on individual rationality and grand coalition rationality
- ◆ Core is too strong, in many cases, it is empty.(no solution)
- ◆ Many solutions satisfy the core solution concept. (multi solutions)
- ◆ Nucleus is in the center of the core vectors.



Transfer Scheme for search the core

- ◆ Repeat

Choose a coalition T

For every agent $i \in T$,

$$x_i^{\text{new}} = x_i + (v_T - \sum_{i \in T} x_i) / |T|$$

and $x_i^{\text{new}} = x_i$ for $i \notin T$

- ◆ Maintain feasibility(???): for every agent $i \in A$,

$$x_i = x_i^{\text{new}} - (\sum_{j \in A} x_j^{\text{new}} - \sum_{S \in CS} v_S) / |A|$$



Shapley Value

Characterize the payoff division problem axiomatically

- ◆ Symmetry: if i and j are interchangeable
then $x_i = x_j$
- ◆ Dummies: if i is dummy, then $x_i = v_{\{i\}}$
- ◆ Additivity: for any two games v and w ,
 x_i in $v+w$ equals x_i in v plus x_i in w
Namely $(v+w)_S = v_S + w_S$

- 
- ◆ Dummy agent: $v_{S \cup \{i\}} - v_S = v_{\{i\}}$ for all S
 - ◆ Symmetry: If agent i and j are interchangeable, then for all S that includes i but not j , $v_{(S \setminus \{i\}) \cup \{j\}} = v_S$



Theorem: Shapley value

- ◆ The only payoff division scheme that satisfies these three axioms

$$x_i = \sum_{S \subseteq A} [v_S - v_{S-\{i\}}] (|A| - |S|)! (|S| - 1)! / |A|!$$

- ◆ Marginal contribution of agent i to the coalition structure, averaged over all possible joining orders.



Advantages and Disadvantages of Shapley Value Scheme

- ◆ Shapley value is unique and always exists
- ◆ Shapley value is Pareto efficient in the sense that entire value of the coalition structure gets distributed among the agents.
- ◆ However, it does not guarantee that all subgroups of agents are better off in the coalition structure than by breaking off into a coalition of their own.
- ◆ The computation of the Shapley value takes $|A|!$
- ◆ Shapley value is member-based payoff division did not take into consideration of effort-fairness.



homework 4 Due June 5

- ◆ Read a paper on coalition or multi-agent payoff division (or profit sharing)
- ◆ Write down the authors and affiliations
- ◆ Write down the source of the paper
- ◆ Show the methods used for payoff division
- ◆ Discuss the issues raised on the payoff division in the paper.